

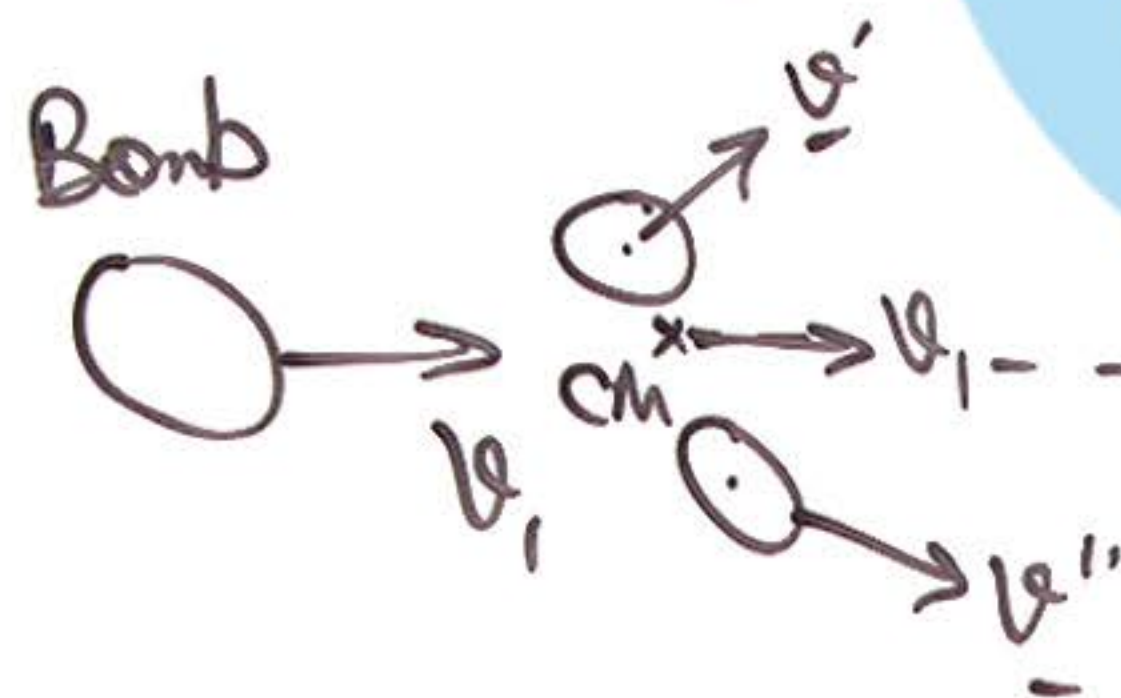
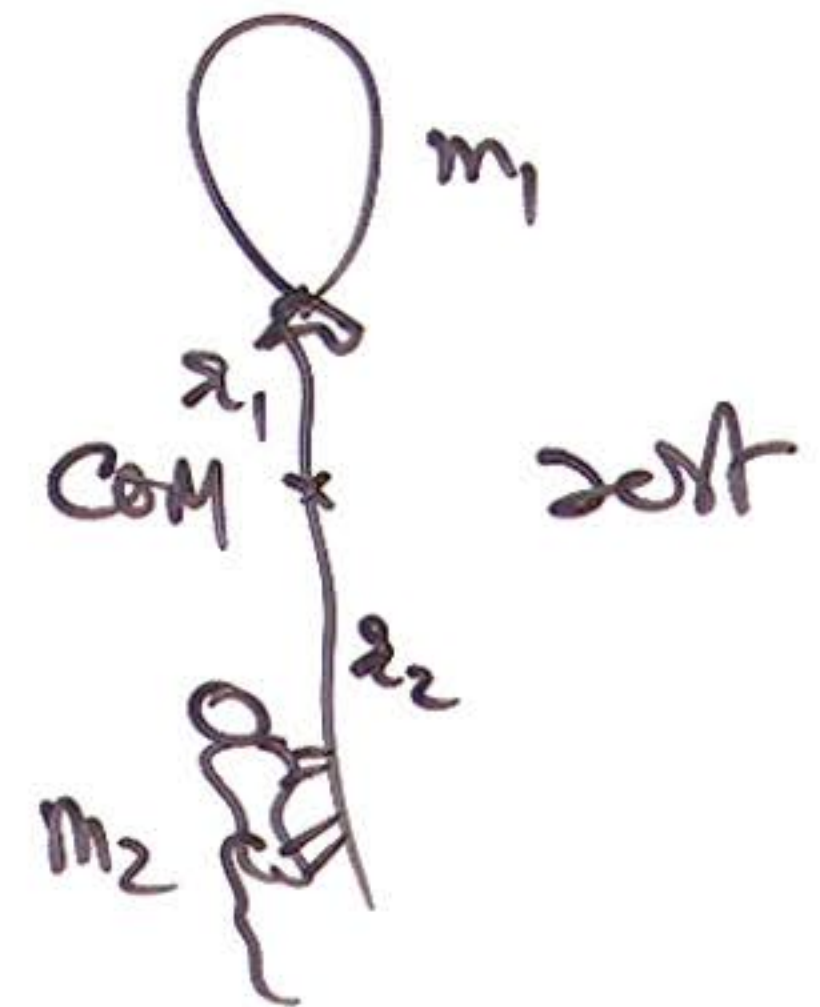
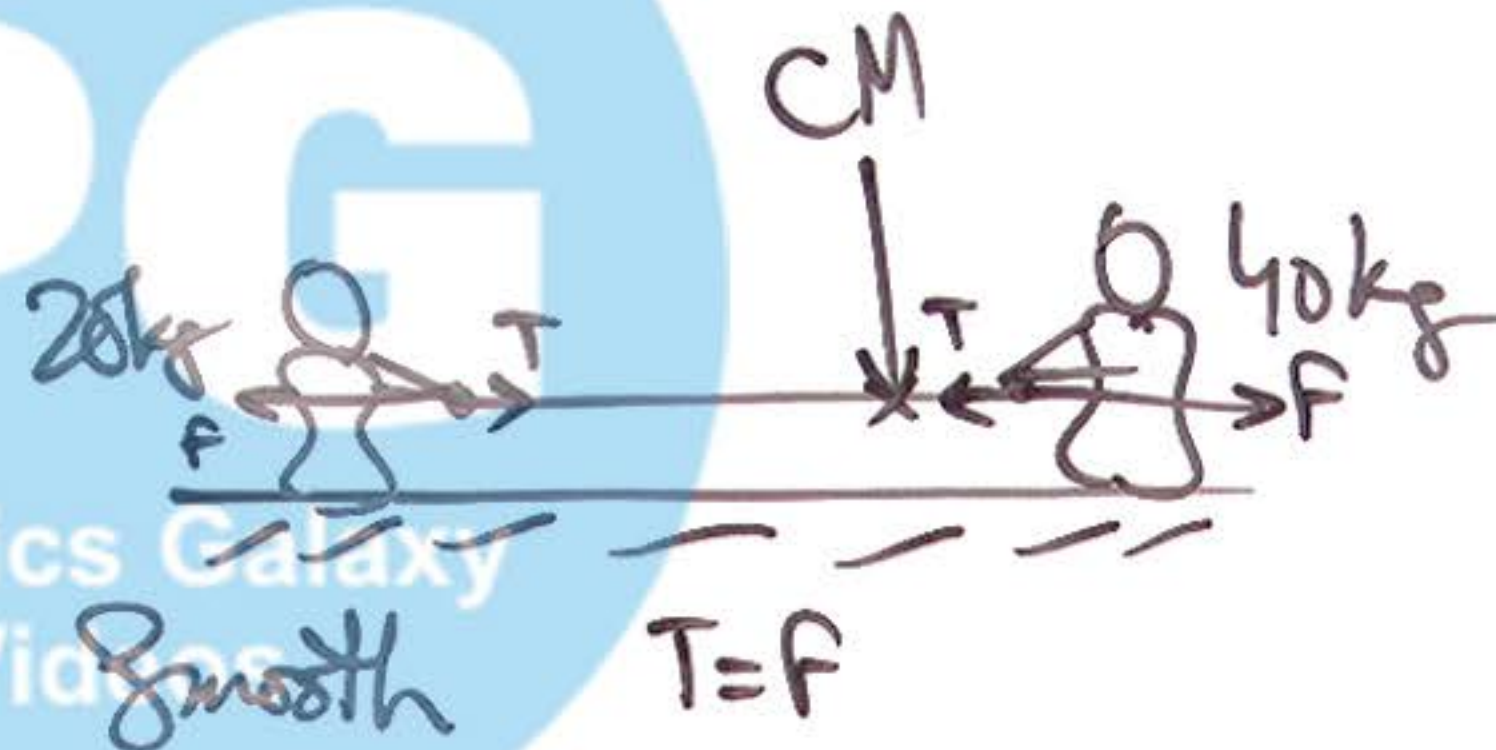
① Conservation of linear momentum:

$$a_{cm} = \frac{\sum \vec{F}_{ext}}{\sum m_i} \quad \& \quad v_{cm} = \frac{\sum \vec{P}_{sys}}{\sum m_i}$$

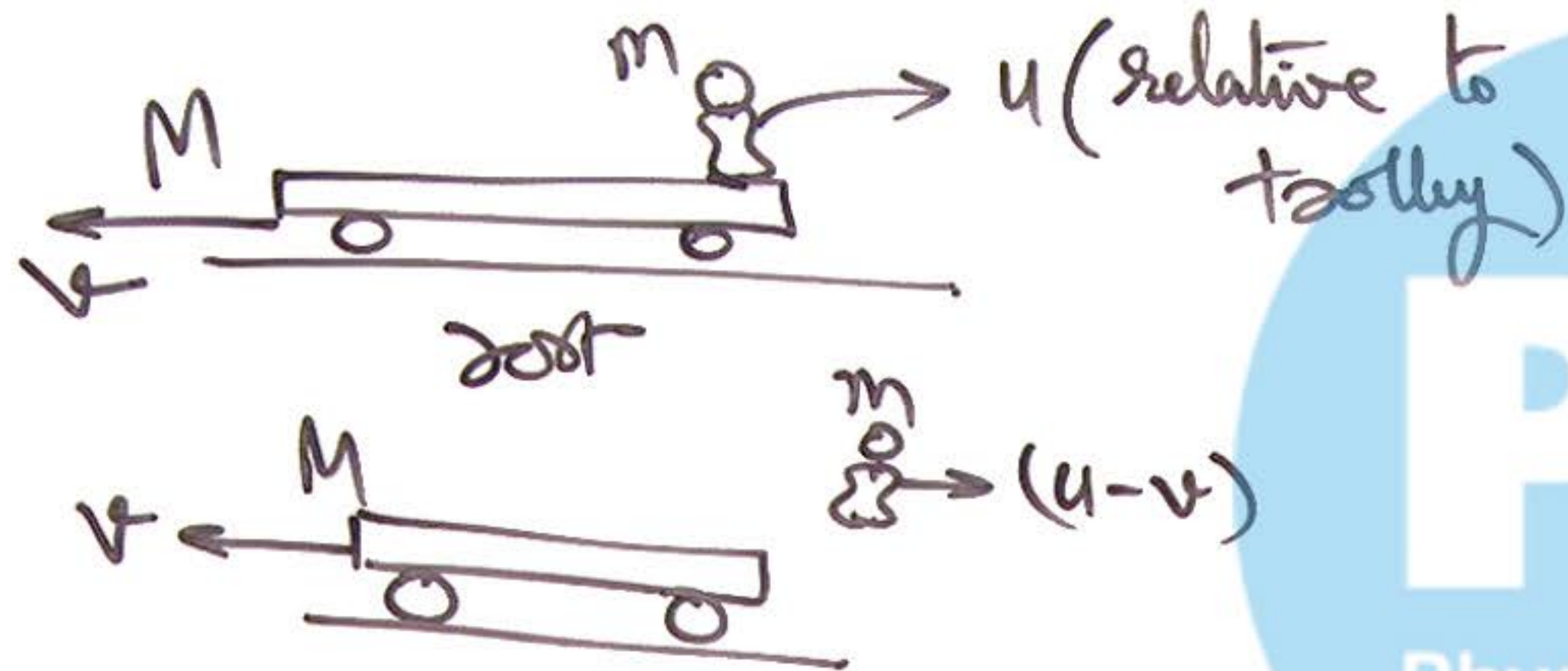
$$\Rightarrow \text{If } F_{ext} = 0 \Rightarrow$$

if $v_{cm} = \text{const} \Rightarrow$ remain const
if $v_{cm} = 0$ then it remain 0.

$$\text{If } \sum \vec{F}_{ext} = 0 \Rightarrow \sum \vec{P}_{sys} = \text{Constant}$$



② Boy-Trolley Cases:



$$Mv = m(u-v)$$

$$v = \frac{mu}{M+m} \checkmark$$



$$Mv = mu \Rightarrow v = \frac{mu}{M} \checkmark$$

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relation of
Always use Cons of momentum
in inertial frame (ground)

③ Concept of Impulse:

$$F = \frac{dp}{dt}$$

$$\Rightarrow dp = F dt$$

$$\xrightarrow{F = \text{const}} \Delta P = F \Delta t$$

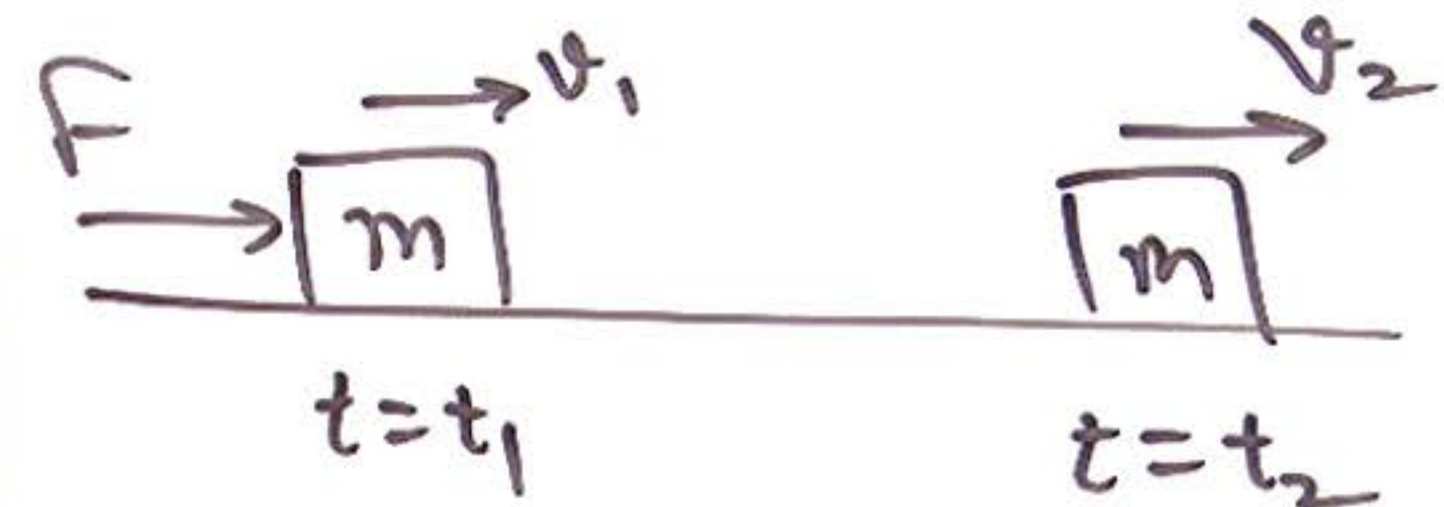
$$\xrightarrow{F = f(t)} \Delta P = \int_{t_1}^{t_2} F dt$$



$$\frac{P-J \text{ eq.}}{P-J \text{ eq.}}$$

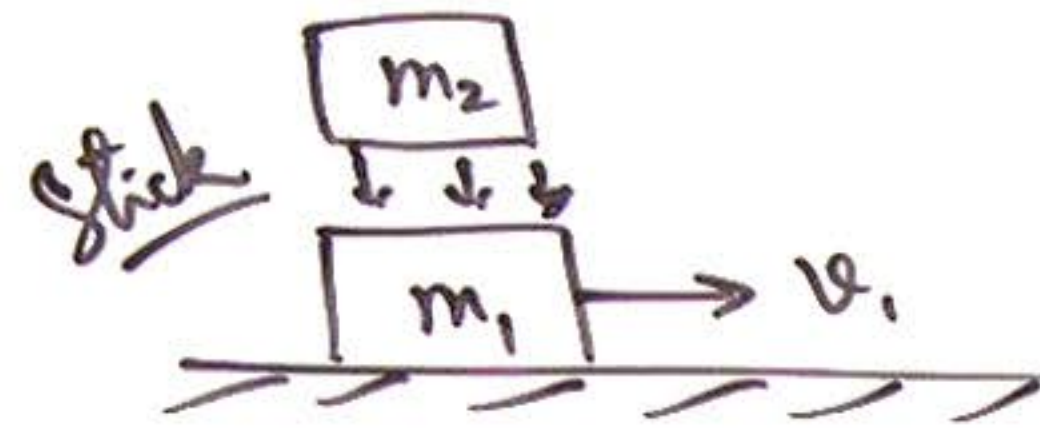
$$\vec{P}_i + \vec{J} = \vec{P}_f$$

$$mv_1 + F(t_2 - t_1) = mv_2$$



$$\therefore F = f(t) \Rightarrow J = \int_{t_1}^{t_2} F dt$$

④ Mass addition on moving bodies:



p_i

p_f

$$m_1 v_1 = (m_1 + m_2) v_f$$

$$v_f = \frac{m_1 v_1}{(m_1 + m_2)}$$

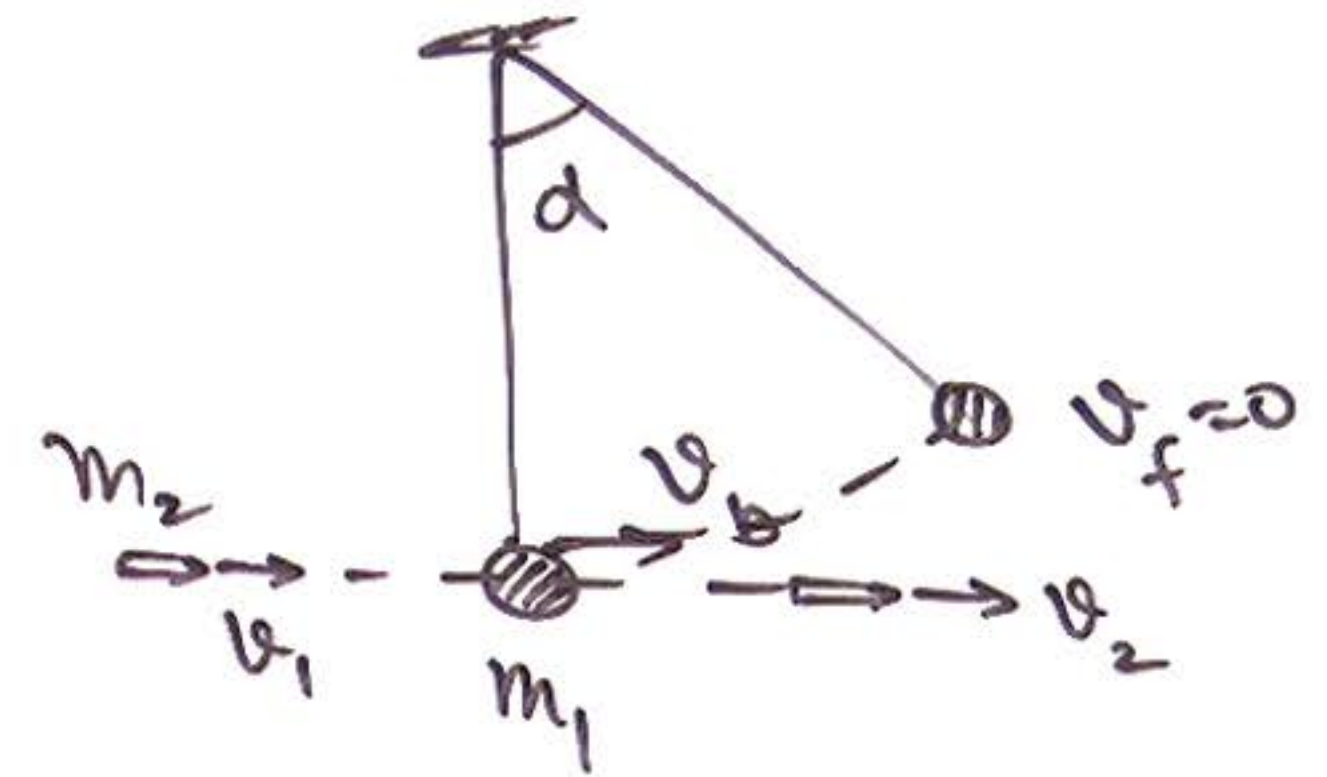


p_i

p_f

$$m_1 v_1 + m_2 v_2 = (m_1 + m_2) v_f$$

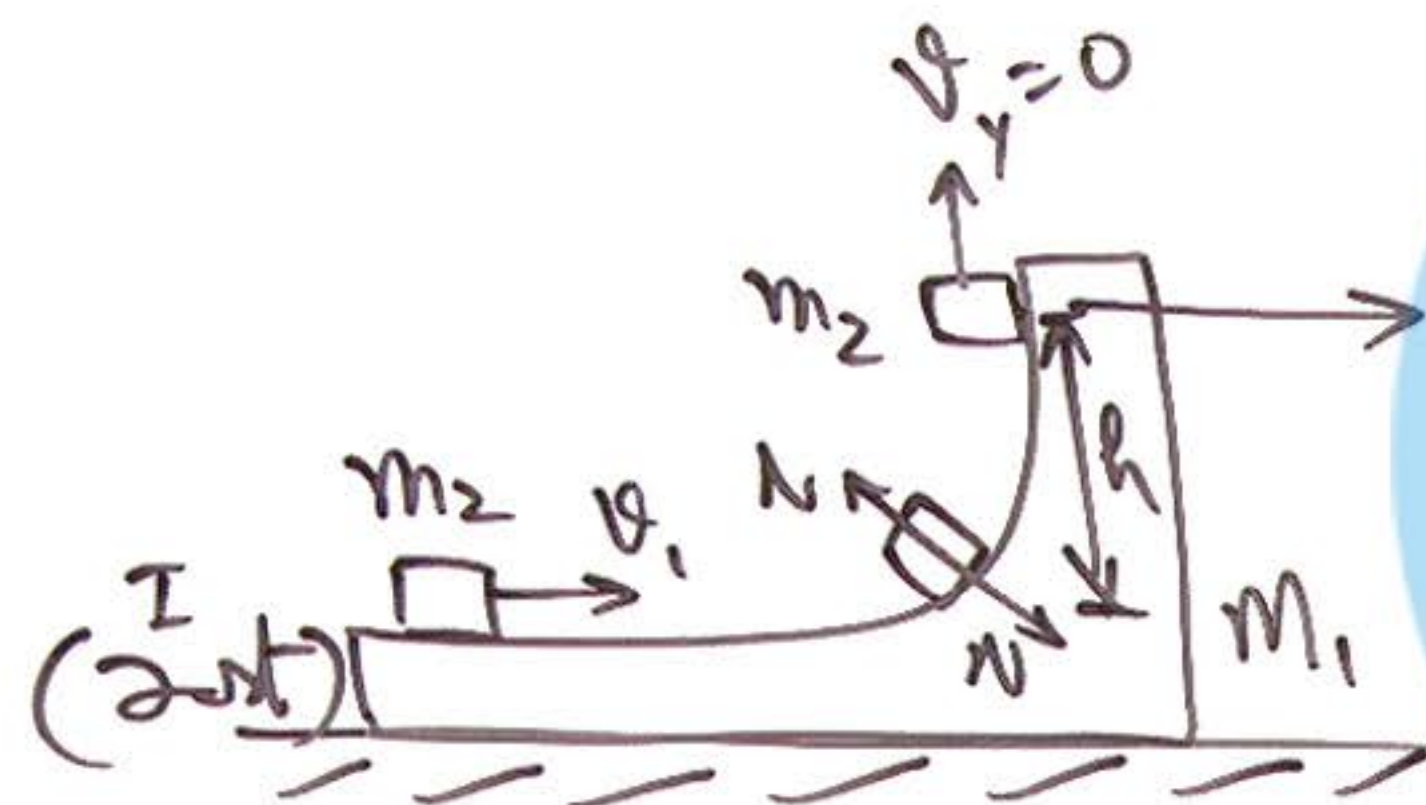
$$v_f = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$



$$m_2 v_1 = m_1 v_b + m_2 v_2$$

$$v_b = -$$

⑤ Block moving over another block cases:



$$m_2 u_1 = (m_1 + m_2) u_f$$

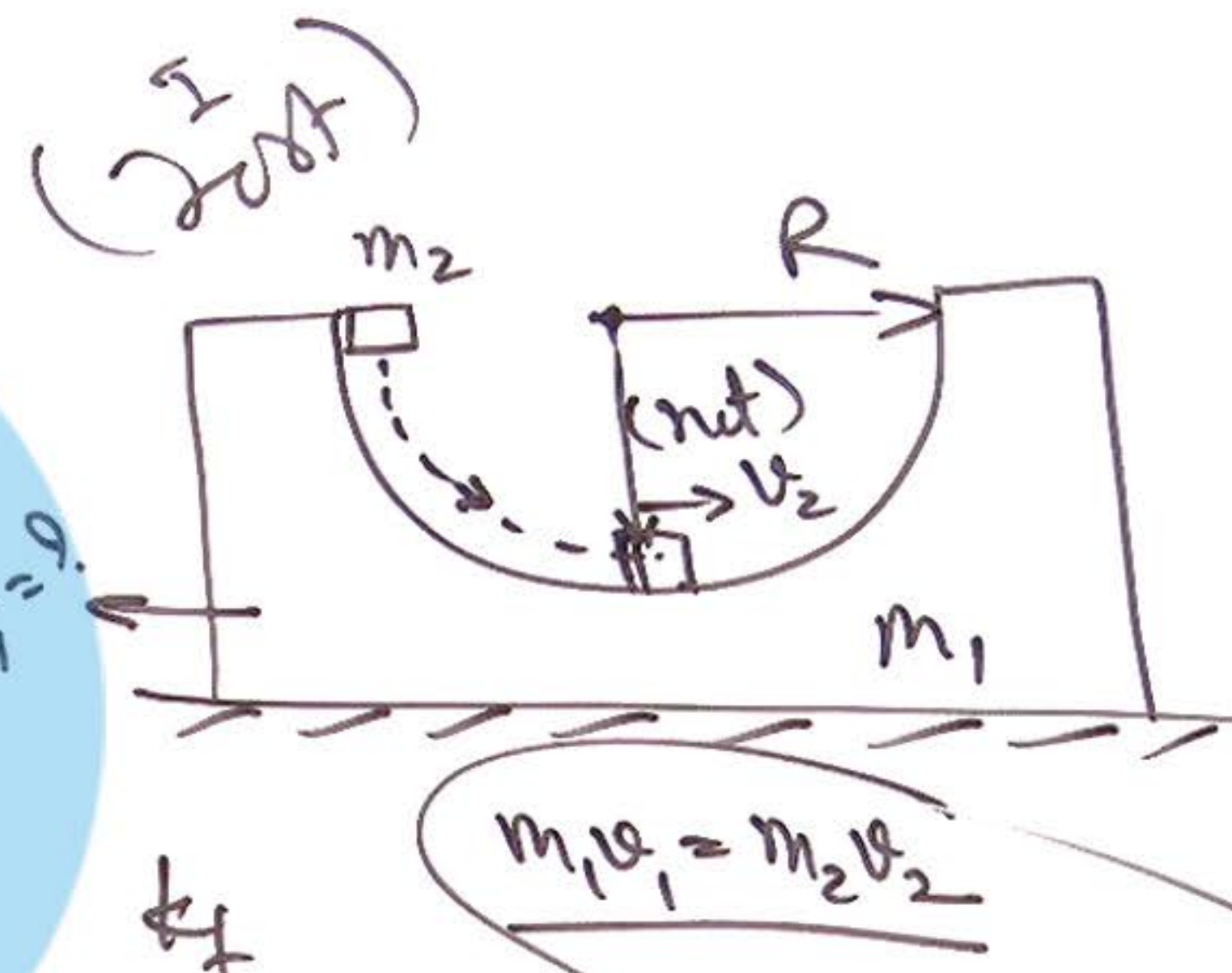
$$u_f = \text{---}$$

$$u_f = ?$$

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$$\frac{1}{2} m_2 u_1^2 - m_2 g h = \frac{1}{2} (m_1 + m_2) u_f^2$$

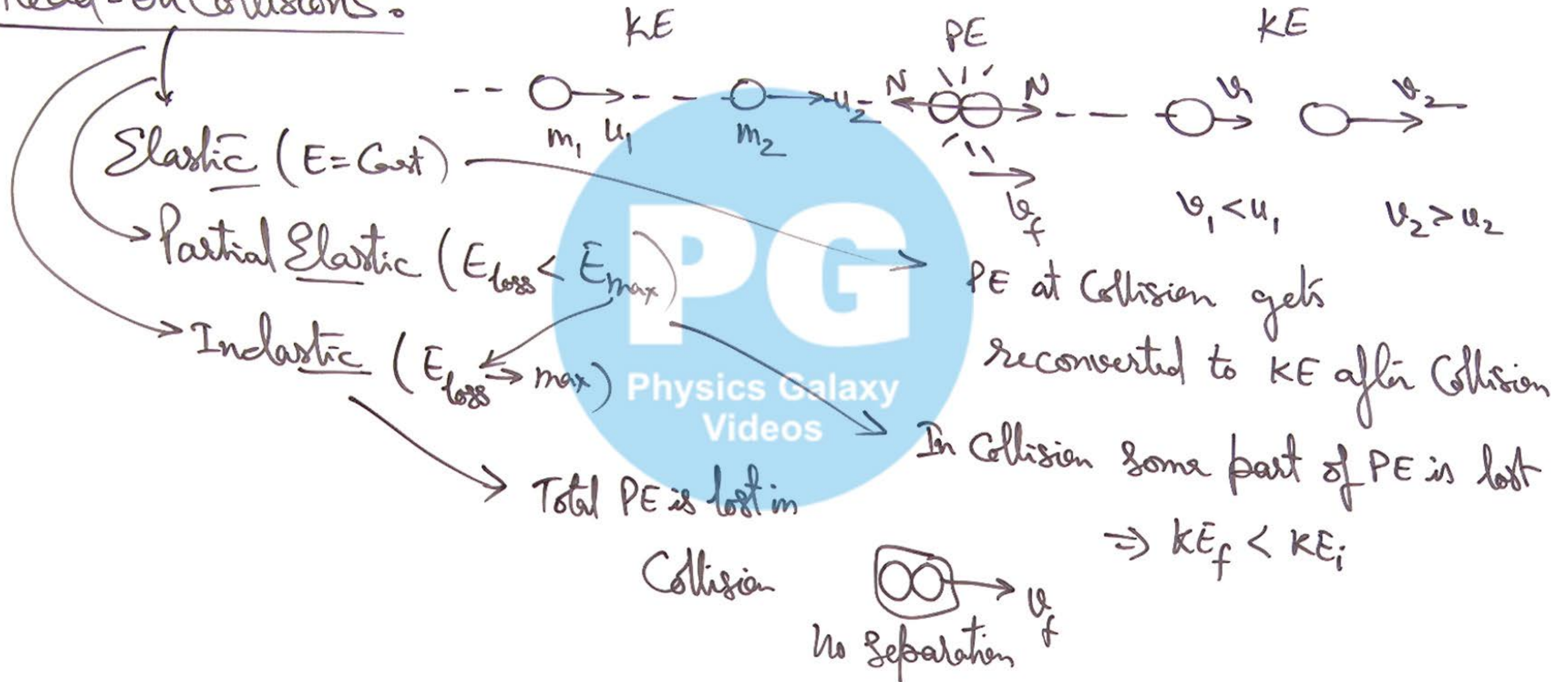
$$h = \text{---}$$



$$m_1 u_1 = m_2 u_2$$

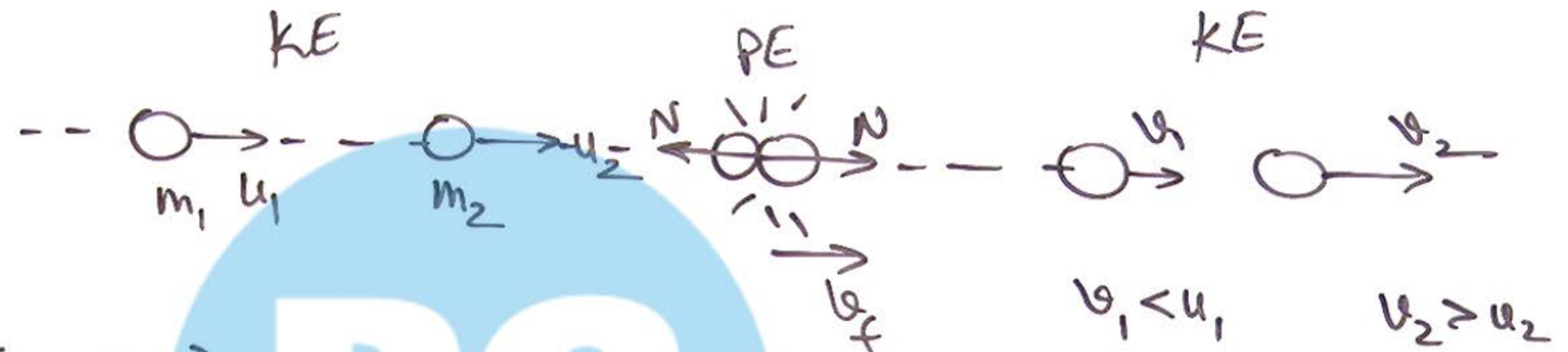
$$m g R = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

⑥ Head-on Collisions:



⑥ Head-on Collisions:

- Elastic ($e=1$)
- Partial Elastic ($0 < e < 1$)
- Inelastic ($e=0$)



$$v_2 - v_1 = e(u_1 - u_2) \quad \text{--- (1)}$$

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \text{--- (2)}$$

Coeff of Restitution

$$e = \left(\frac{\text{sep vel}}{\text{app vel}} \right) = \left(\frac{v_2 - v_1}{u_1 - u_2} \right)$$

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2} \right) u_1 + \frac{m_2(1+e)}{m_1 + m_2} \cdot u_2$$

$$v_2 = \left(\frac{m_2 - em_1}{m_1 + m_2} \right) u_2 + \frac{m_1(1+e)}{m_1 + m_2} \cdot u_1$$

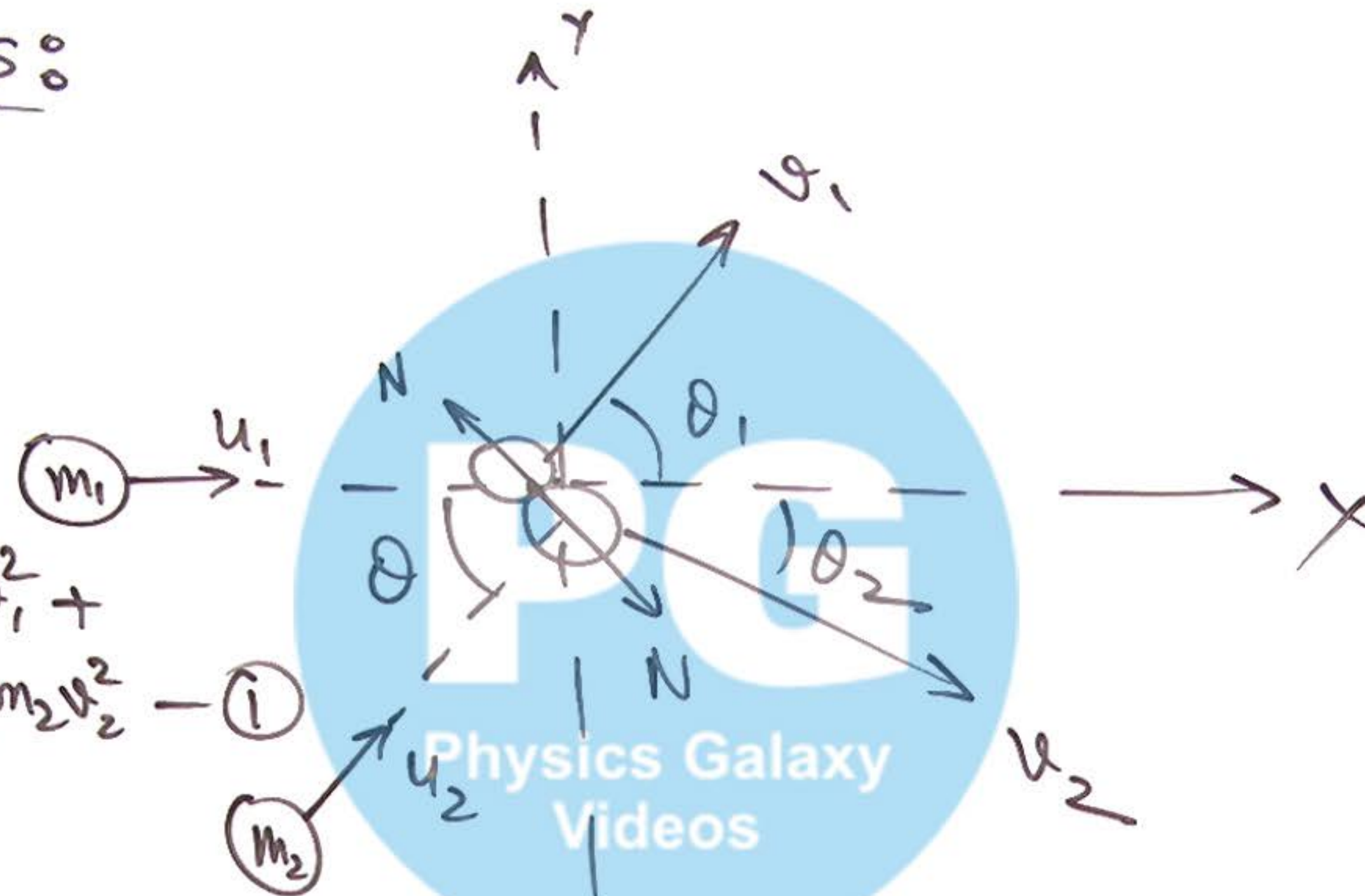
⑦ Oblique Collisions: (2D Collision)

for elastic collision

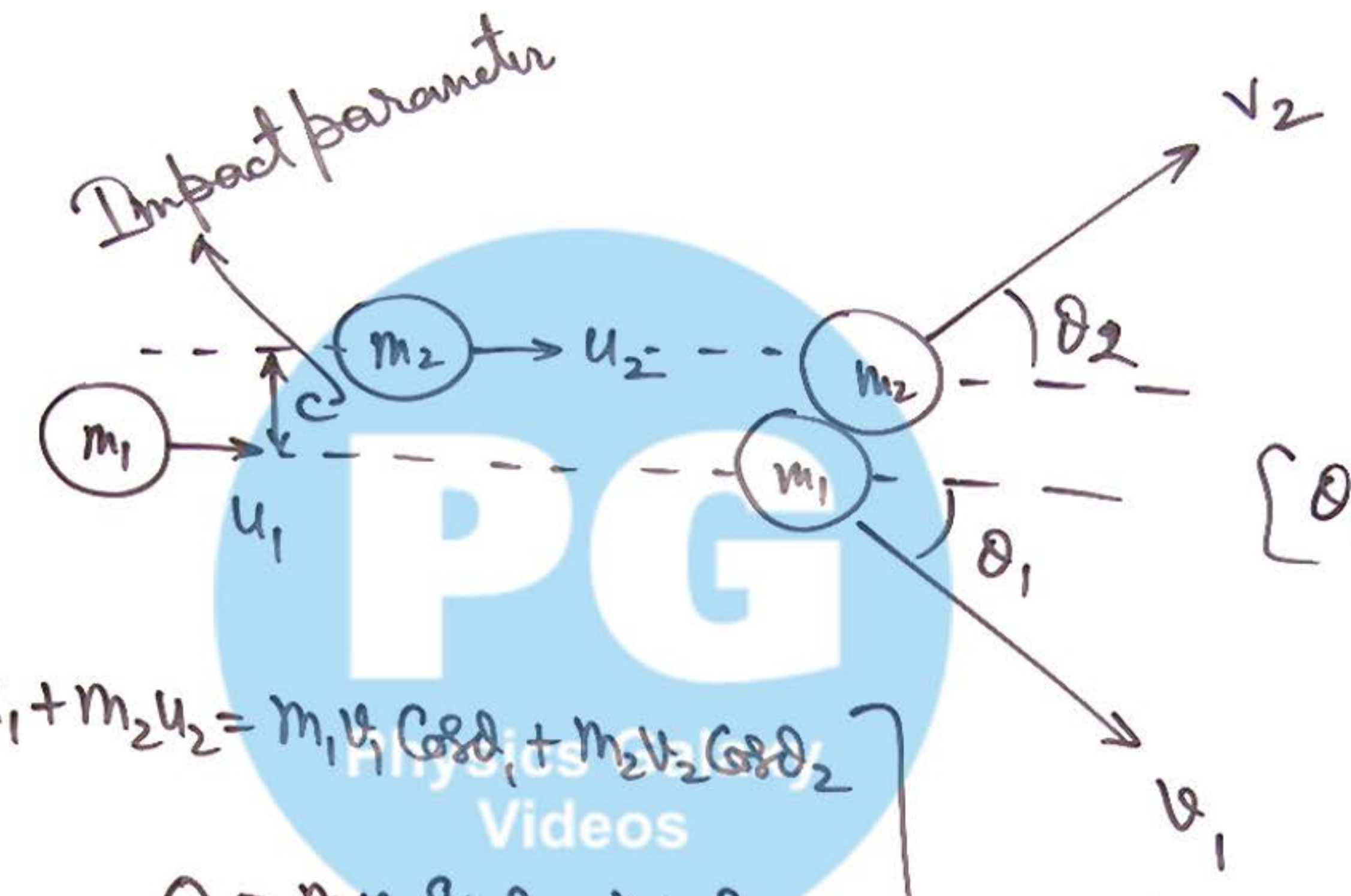
$$\frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2 \quad \text{--- (1)}$$

$$m_1u_1 + m_2u_2\cos\theta = m_1v_1\cos\theta_1 + m_2v_2\cos\theta_2 \quad \text{--- (2)}$$

$$m_2u_2\sin\theta = m_1v_1\sin\theta_1 - m_2v_2\sin\theta_2 \quad \text{--- (3)}$$



⑦ Oblique Collisions: (2D Collision)



Impact parameter

m_1 u_1 m_2 u_2 m_1 m_2 v_1 v_2 θ_1 θ_2

$[\theta_1 + \theta_2] \rightarrow$ Angle of Divergence.

$$m_1 u_1 + m_2 u_2 = m_1 v_1 \cos \theta_1 + m_2 v_2 \cos \theta_2$$

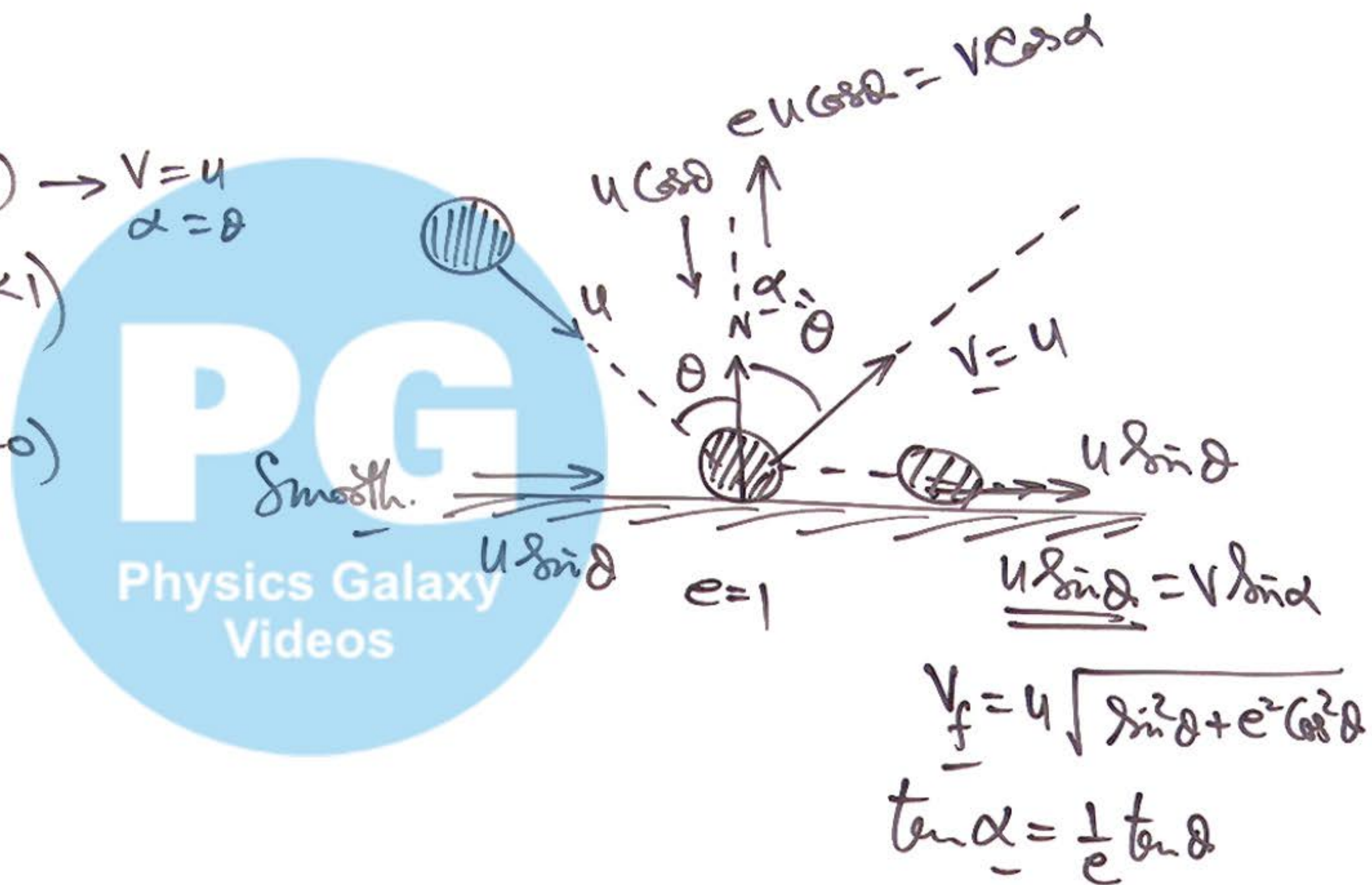
$$0 = m_2 v_2 \sin \theta_2 - m_1 v_1 \sin \theta_1$$

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2$$

$\rightarrow$ If $u_2 = 0$ & $m_1 = m_2$ $\theta_1 + \theta_2 = 90^\circ$
 Then always we get

⑧ Collision with a floor:

- elastic ($e=1$) → $V=U$
 $\alpha=\theta$
- p. elastic ($0 < e < 1$)
- inelastic ($e=0$)



⑨ Cases of Mass Variation:

at $t=0$

M

v

u (rel to rocket)

$t \rightarrow m$

$(M - rt)$

$ru = m \frac{dv}{dt}$

Thrust on Rocket

Mass ejection rate = $\underline{r}$ kg/s

$F_{th} = (ru)$

$\int_0^t \frac{dt}{M - rt} = \int_0^v \frac{dv}{ru}$

$v = -u \ln\left(\frac{M - rt}{M}\right)$

$\Rightarrow v = u \ln\left(\frac{M}{M - rt}\right)$

$\frac{v}{ru} = -\frac{1}{r} \left[\ln(M - rt) \right]_0^t$